

PS-04 — your data model

PackagePro — Dynamic Tour Packages

Kognivera Hackathon 2026 · Travel & Tourism · data model v1.1.0-rc1

The problem statement itself, the 24-hour MVP scope and the XR device requirement live in the hackathon application, on your statement's page. This document is the data you have been given to build it with: every table, every field, and what each one is for.

You have **28,103 rows across 21 tables**. 20 of them are the tables this statement is built on; the remaining 1 is a reference table the others point at, included so the database works on its own.

All of it is in the `data/` folder beside this document: as `PS-04.db` (SQLite, indexed, ready to query), as CSV, and as DDL for Postgres and SQLite.

What the data gives you

60 curated packages decomposed into 420 **individually swappable** components with signed price deltas, plus 120 guides with languages, specialisations and a 30-day availability calendar.

Watch out for this one

`price_delta` is signed and can be negative. Reprice as base + the deltas you keep, in Decimal — a float here produces a total that is 0.01 off and impossible to explain.

The tables this statement is built on

TABLE	ROWS	WHAT YOU USE IT FOR
<code>bookings</code>	1,996	The order header — where inventory, payment and itinerary meet. Idempotency key is mandatory, not optional.
<code>cities</code>	60	The geographic anchor of the whole model. 60 cities; every hotel, POI, package, advisory and weather row hangs off one.
<code>countries</code>	30	ISO country reference. Every city, currency default and calling code resolves here.
<code>currencies</code>	25	carries the true minor-unit exponent so JPY/KWD display correctly even though storage is always DECIMAL(12,2).
<code>hotels</code>	300	Fewer, richer properties. Depth (reviews, room types, media) matters more than catalogue size —.
<code>itineraries</code>	803	A versioned plan belonging to a trip. version is what makes PS-11's conflict handling tractable.

TABLE	ROWS	WHAT YOU USE IT FOR
itinerary_items	8,583	The atom of the portal, and the single most-shared object across the thirteen builds. If a team implements one shared shape, this is it.
languages	26	Rule R6: BCP-47 is the only legal way to say 'language' anywhere in the model.
package_components	420	the swappable line. You cannot swap an item inside a pipe-separated string, and swapping is PS-04's core requirement.
tour_packages	60	Curated catalogues are small in real life. The depth lives in package_components.
trips	600	The container that gives dates, party, destination and budget to everything else. Seven statements produce or consume one.
user_preferences	1,200	Explicit preference signal. PS-04's language-preference requirement reads from here.
users	1,200	The traveller identity every personalisation hangs off. Segmented heavy / light / cold_start so APS-04 can prove cold start.
amenities	60	Replaces the pipe-separated amenity string. Filters in PS-02 join here.
categories	70	One two-level taxonomy shared by POI type, package theme and expense category — so the three never drift apart.
guide_availability	3,600	Makes 'is this guide free on the 12th?' a real query rather than an assumption.
hotel_media	1,500	Image references with role and alt text. PS-05 uses hero/room rows as the fallback when no XR scene exists.
price_history	5,950	Computed price over time with the driving factors alongside — the shape APS-02's explainability requirement has to produce.
tour_guides	120	PS-04's added guide dimension — selectable by language, specialisation, availability and price.
transfers	300	Airport and intercity legs with cost, duration and carbon — the non-POI edges in an optimised day.

Reference tables, included so the database is valid

You will mostly join through these rather than think about them.

TABLE	ROWS	WHAT IT IS
hotel_room_types	1,200	The bookable unit. APS-05's no-oversell guarantee is defended at this grain.

How they fit together

Open `02_DATA_MODEL_DIAGRAM.html` in a browser for the clickable version — it shows these tables and nothing else. Download it first; it will not render inside SharePoint.

Some tables point at "any bookable thing" using an `(entity_type, entity_id)` pair rather than a typed foreign key. That is deliberate: it is what lets one feature refer to a hotel, a flight, a point of interest or a

package without a separate join table for each. The legal values of `entity_type` are in `data/enums.json`.

Every field, table by table

Columns marked **PK** are the primary key. **FK** shows what a column points at. Enum columns list their legal values — anything else is rejected by the conformance check.

amenities

Reference & geography · 60 rows · IDs start `amn_`

Replaces the pipe-separated amenity string. Filters in PS-02 join here.

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>amenity_id</code>	text	PK	<code>amn_</code> prefixed.
<code>code</code>	text	UNIQUE · NOT NULL	<code>snake_case</code> machine code, e.g. <code>free_wifi</code> .
<code>label</code>	text	NOT NULL	Display label.
<code>amenity_group</code>	text	NOT NULL · one of <code>connectivity</code> , <code>wellness</code> , <code>food_beverage</code> , <code>family</code> , <code>accessibility</code> , <code>transport</code> , <code>business</code> , <code>outdoor</code>	
<code>icon_hint</code>	text		Suggested icon name, purely advisory.
<code>updated_at</code>	timestampz	NOT NULL	

categories

Reference & geography · 70 rows · IDs start `cat_`

One two-level taxonomy shared by POI type, package theme and expense category — so the three never drift apart.

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>category_id</code>	text	PK	<code>cat_</code> prefixed.
<code>code</code>	text	UNIQUE · NOT NULL	<code>snake_case</code> .
<code>label</code>	text	NOT NULL	
<code>parent_category_id</code>	text	FK → <code>categories.category_id</code>	Null for top-level; self-referencing.
<code>applies_to</code>	text	NOT NULL	<code>poi</code>
<code>updated_at</code>	timestampz	NOT NULL	

currencies

Reference & geography · 25 rows · IDs start `cur_`

carries the true minor-unit exponent so JPY/KWD display correctly even though storage is always DECIMAL(12,2).

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>currency_id</code>	text	PK	<code>cur_</code> prefixed.
<code>iso4217</code>	char(3)	UNIQUE · NOT NULL	e.g. INR.
<code>name</code>	text	NOT NULL	
<code>symbol</code>	text	NOT NULL	
<code>minor_unit_exponent</code>	smallint	NOT NULL	0 for JPY/KRW, 2 default, 3 for KWD/BHD.
<code>display_locale</code>	text	NOT NULL	BCP-47 locale used for formatting.
<code>updated_at</code>	timestampz	NOT NULL	

languages

Reference & geography · 26 rows · IDs start `lng_`

Rule R6: BCP-47 is the only legal way to say 'language' anywhere in the model.

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>language_id</code>	text	PK	<code>lng_</code> prefixed.
<code>bcp47</code>	text	UNIQUE · NOT NULL	e.g. ta, hi, en-IN — never 'Tamil'.
<code>english_name</code>	text	NOT NULL	
<code>native_name</code>	text	NOT NULL	
<code>script</code>	text	NOT NULL	ISO-15924, e.g. Taml, Deva, Latn.
<code>rtl</code>	bool	NOT NULL	Right-to-left rendering flag.
<code>tts_supported</code>	bool	NOT NULL	Relevant to PS-13 voice output.
<code>updated_at</code>	timestampz	NOT NULL	

price_history

Availability & pricing · 5,950 rows · IDs start `phs_`

Computed price over time with the driving factors alongside — the shape APS-02's explainability requirement has to produce.

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>history_id</code>	text	PK	<code>phs_</code> prefixed.

COLUMN	TYPE	CONSTRAINTS	NOTES
entity_type	text	NOT NULL · one of room_type, flight_fare, guide, poi	
entity_id	text	NOT NULL	
effective_date	date	NOT NULL	
price	decimal(12,2)	NOT NULL	D3.
currency	char(3)	FK → currencies.iso4217 · NOT NULL	
baseline_price	decimal(12,2)	NOT NULL	
demand_index	decimal(6,3)	NOT NULL	Forecast output that drove the move.
occupancy_pct	decimal(5,2)	NOT NULL	
lead_time_factor	decimal(6,3)	NOT NULL	
seasonality_factor	decimal(6,3)	NOT NULL	
event_factor	decimal(6,3)	NOT NULL	Uplift from a festival in events_festivals.
competitor_factor	decimal(6,3)	NOT NULL	
bound_clamped	bool	NOT NULL	True where price_bounds clipped the computed price.
explanation	text	NOT NULL	Human-readable factor summary.
computed_at	timestamptz	NOT NULL	

Unique together: (entity_type, entity_id, effective_date)

countries

Reference & geography · 30 rows · IDs start cnt_

ISO country reference. Every city, currency default and calling code resolves here.

COLUMN	TYPE	CONSTRAINTS	NOTES
country_id	text	PK	Canonical ID, cnt_ prefixed.
iso2	char(2)	UNIQUE · NOT NULL	ISO-3166-1 alpha-2, e.g. IN.
iso3	char(3)	UNIQUE · NOT NULL	ISO-3166-1 alpha-3, e.g. IND.
name	text	NOT NULL	English short name.
default_currency	char(3)	FK → currencies.iso4217 · NOT NULL	ISO-4217 code.

COLUMN	TYPE	CONSTRAINTS	NOTES
calling_code	text	NOT NULL	E.164 country calling code, e.g. +91.
region	text	NOT NULL	UN sub-region grouping.
updated_at	timestampz	NOT NULL	Rule R4: UTC, ISO-8601 with offset.

cities

Reference & geography · 60 rows · IDs start `cty_`

The geographic anchor of the whole model. 60 cities; every hotel, POI, package, advisory and weather row hangs off one.

COLUMN	TYPE	CONSTRAINTS	NOTES
city_id	text	PK	cty_ prefixed.
name	text	NOT NULL	City name.
state	text		State / province, nullable for city-states.
country_id	text	FK → countries.country_id · NOT NULL	
country_code	char(2)	NOT NULL	Denormalised ISO2 for convenient joins.
lat	decimal(9,6)	NOT NULL	Rule R7: WGS-84, 6dp.
lng	decimal(9,6)	NOT NULL	Rule R7: WGS-84, 6dp.
timezone	text	NOT NULL	IANA zone, e.g. Asia/Kolkata.
region	text	NOT NULL	Domestic region grouping, e.g. South India.
population	int		Approximate, for demand weighting.
season_profile	text	NOT NULL · one of winter, summer, monsoon, post_monsoon, spring, autumn	Dominant season at the peak travel window.
peak_months	text	NOT NULL	Comma-separated month numbers, e.g. 10,11,12.
primary_language	text	FK → languages.bcp47 · NOT NULL	Rule R6: BCP-47 tag.
description	text		One-paragraph orientation blurb, used by PS-13.

COLUMN	TYPE	CONSTRAINTS	NOTES
status	text	NOT NULL · one of <code>active</code> , <code>inactive</code> , <code>archived</code> , <code>draft</code>	
updated_at	timestampz	NOT NULL	

hotels

Supply & catalogue · 300 rows · IDs start `htl_`

Fewer, richer properties. Depth (reviews, room types, media) matters more than catalogue size —.

COLUMN	TYPE	CONSTRAINTS	NOTES
hotel_id	text	PK	htl_ prefixed.
city_id	text	FK → <code>cities.city_id</code> · NOT NULL	
name	text	NOT NULL	~3% near-duplicate names injected deliberately.
property_type	text	NOT NULL · one of <code>hotel</code> , <code>resort</code> , <code>homestay</code> , <code>hostel</code> , <code>apartment</code> , <code>boutique</code> , <code>heritage</code> , <code>guesthouse</code>	
star_rating	smallint	NOT NULL	1–5, CHECK constrained.
guest_score	decimal(2,1)		0.0–10.0; null for a handful of new properties.
review_count	int	NOT NULL	Denormalised count, must agree with <code>hotel_reviews</code> .
address_line	text	NOT NULL	
lat	decimal(9,6)	NOT NULL	
lng	decimal(9,6)	NOT NULL	
distance_to_centre_km	decimal(6,2)	NOT NULL	PS-02 filter: distance to a landmark.
description	text	NOT NULL	Plausible prose, embeddable.
base_currency	char(3)	FK → <code>currencies.iso4217</code> · NOT NULL	
checkin_time	text	NOT NULL	Local HH:MM at the property.
checkout_time	text	NOT NULL	
chain_code	text		Null for independents.

COLUMN	TYPE	CONSTRAINTS	NOTES
has_xr_scene	bool	NOT NULL	PS-05 / APS-07 — does an immersive preview exist.
status	text	NOT NULL · one of active, inactive, archived, draft	
created_at	timestampz	NOT NULL	
updated_at	timestampz	NOT NULL	

tour_guides

Supply & catalogue · 120 rows · IDs start gid_

PS-04's added guide dimension — selectable by language, specialisation, availability and price.

COLUMN	TYPE	CONSTRAINTS	NOTES
guide_id	text	PK	gid_ prefixed.
city_id	text	FK → cities.city_id · NOT NULL	
display_name	text	NOT NULL	Synthetic.
languages	text	NOT NULL	Comma-separated BCP-47 — the PS-04 filter.
specialisation	text	NOT NULL · one of heritage, food, trekking, wildlife, photography, religious, shopping, accessibility	
secondary_specialisation	text	one of heritage, food, trekking, wildlife, photography, religious, shopping, accessibility	
years_experience	smallint	NOT NULL	
rating	decimal(2,1)		Null for new guides.
review_count	int	NOT NULL	
day_rate	decimal(12,2)	NOT NULL	D3.
half_day_rate	decimal(12,2)	NOT NULL	
currency	char(3)	FK → currencies.iso4217 · NOT NULL	
certified	bool	NOT NULL	
bio	text	NOT NULL	Plausible prose.
status	text	NOT NULL · one of active, inactive, archived, draft	

COLUMN	TYPE	CONSTRAINTS	NOTES
updated_at	timestamptz	NOT NULL	

tour_packages

Supply & catalogue · 60 rows · IDs start `pkg_`

Curated catalogues are small in real life. The depth lives in package_components.

COLUMN	TYPE	CONSTRAINTS	NOTES
package_id	text	PK	pkg_ prefixed.
city_id	text	FK → cities.city_id · NOT NULL	Primary destination.
name	text	NOT NULL	
theme	text	NOT NULL · one of adventure, honeymoon, pilgrimage, family, heritage, wellness, wildlife, food_trail	
tier	text	NOT NULL · one of standard, deluxe, premium	
duration_days	smallint	NOT NULL	
duration_nights	smallint	NOT NULL	
base_price	decimal(12,2)	NOT NULL	the figure PS-04 reprices live.
currency	char(3)	FK → currencies.iso4217 · NOT NULL	
min_group_size	smallint	NOT NULL	
max_group_size	smallint	NOT NULL	
difficulty	text	NOT NULL	easy
languages_offered	text	NOT NULL	Comma-separated BCP-47 — PS-04 filters on this.
inclusions	text	NOT NULL	Prose summary.
exclusions	text	NOT NULL	
description	text	NOT NULL	
status	text	NOT NULL · one of active, inactive, archived, draft	
updated_at	timestamptz	NOT NULL	

transfers

Supply & catalogue · 300 rows · IDs start `trf_`

Airport and intercity legs with cost, duration and carbon — the non-POI edges in an optimised day.

COLUMN	TYPE	CONSTRAINTS	NOTES
transfer_id	text	PK	trf_ prefixed.
city_id	text	FK → cities.city_id · NOT NULL	
from_label	text	NOT NULL	
to_label	text	NOT NULL	
mode	text	NOT NULL · one of walk, cycle, auto_rickshaw, cab, bus, metro, train, ferry, flight	
duration_minutes	int	NOT NULL	
distance_km	decimal(7,3)	NOT NULL	
cost	decimal(12,2)	NOT NULL	
currency	char(3)	FK → currencies.iso4217 · NOT NULL	
carbon_kg	decimal(8,3)	NOT NULL	
capacity_pax	smallint	NOT NULL	
accessible	bool	NOT NULL	
status	text	NOT NULL · one of active, inactive, archived, draft	
updated_at	timestamptz	NOT NULL	

users

Identity & preference · 1,200 rows · IDs start usr_

The traveller identity every personalisation hangs off. Segmented heavy / light / cold_start so APS-04 can prove cold start.

COLUMN	TYPE	CONSTRAINTS	NOTES
user_id	text	PK	usr_ prefixed.
display_name	text	NOT NULL	Synthetic — no real people (content policy).
email	text	UNIQUE · NOT NULL	Synthetic @example.invalid addresses only.
home_city_id	text	FK → cities.city_id · NOT NULL	
home_currency	char(3)	FK → currencies.iso4217 · NOT NULL	
locale	text	FK → languages.bcp47 · NOT NULL	UI language, BCP-47.

COLUMN	TYPE	CONSTRAINTS	NOTES
budget_band	text	NOT NULL · one of shoestring, value, mid, premium, luxury	
travel_style	text	NOT NULL · one of budget, comfort, luxury, adventure, slow, cultural, wellness	
traveller_type	text	NOT NULL · one of solo, couple, family, business, friends, senior, backpacker	
segment	text	NOT NULL · one of heavy, light, cold_start	heavy / light / cold_start cohorts.
date_of_signup	date	NOT NULL	
loyalty_tier	text		none
status	text	NOT NULL · one of active, inactive, archived, draft	
created_at	timestamptz	NOT NULL	
updated_at	timestamptz	NOT NULL	

guide_availability

Supply & catalogue · 3,600 rows · IDs start gav_

Makes 'is this guide free on the 12th?' a real query rather than an assumption.

COLUMN	TYPE	CONSTRAINTS	NOTES
availability_id	text	PK	gav_ prefixed.
guide_id	text	FK → tour_guides.guide_id · NOT NULL	
for_date	date	NOT NULL	
is_available	bool	NOT NULL	
slots_available	smallint	NOT NULL	0, 1 or 2 half-day slots.
price_multiplier	decimal(4,2)	NOT NULL	Peak-date uplift, e.g. 1.25.
updated_at	timestamptz	NOT NULL	

Unique together: (guide_id, for_date)

hotel_room_types

Supply & catalogue · 1,200 rows · IDs start rmt_ · reference table

The bookable unit. APS-05's no-oversell guarantee is defended at this grain.

COLUMN	TYPE	CONSTRAINTS	NOTES
room_type_id	text	PK	rmt_ prefixed.
hotel_id	text	FK → hotels.hotel_id · NOT NULL	
name	text	NOT NULL	e.g. Deluxe Garden View.
max_occupancy	smallint	NOT NULL	
max_adults	smallint	NOT NULL	
max_children	smallint	NOT NULL	
bed_config	text	NOT NULL · one of single, twin, double, queen, king, bunk, twin_double	
size_sqm	smallint		
base_rate	decimal(12,2)	NOT NULL	NUMERIC, never FLOAT.
currency	char(3)	FK → currencies.iso4217 · NOT NULL	always paired with the amount.
total_units	int	NOT NULL	Deliberately scarce on some rows so APS-05 has contention.
smoking_allowed	bool	NOT NULL	
status	text	NOT NULL · one of active, inactive, archived, draft	
updated_at	timestampz	NOT NULL	

package_components

Supply & catalogue · 420 rows · IDs start pcm_

the swappable line. You cannot swap an item inside a pipe-separated string, and swapping is PS-04's core requirement.

COLUMN	TYPE	CONSTRAINTS	NOTES
component_id	text	PK	pcm_ prefixed.
package_id	text	FK → tour_packages.package_id · NOT NULL	
component_type	text	NOT NULL · one of hotel, flight, poi, transfer, guide, meal, insurance, entry_ticket	
entity_type	text	one of hotel, room_type, rate_plan, flight, flight_fare, poi, package, package_component, guide, transfer, event, xr_scene	Polymorphic pointer — see the supply reference spine.

COLUMN	TYPE	CONSTRAINTS	NOTES
entity_id	text		Canonical ID of the referenced supply row.
day_index	smallint	NOT NULL	1-based day within the package.
slot	text	NOT NULL	morning
title	text	NOT NULL	
quantity	smallint	NOT NULL	
price_delta	decimal(12,2)	NOT NULL	signed; what swapping this line does to the total.
currency	char(3)	FK → currencies.iso4217 · NOT NULL	
is_optional	bool	NOT NULL	
is_swappable	bool	NOT NULL	
swap_group	text		Components sharing a swap_group are alternatives for each other.
updated_at	timestamptz	NOT NULL	

trips

Trip & itinerary · 600 rows · IDs start `trp_`

The container that gives dates, party, destination and budget to everything else. Seven statements produce or consume one.

COLUMN	TYPE	CONSTRAINTS	NOTES
trip_id	text	PK	trp_ prefixed.
owner_user_id	text	FK → users.user_id · NOT NULL	
title	text	NOT NULL	
origin_city_id	text	FK → cities.city_id	
destination_city_id	text	FK → cities.city_id · NOT NULL	
start_date	date	NOT NULL	Rule R4: zoneless calendar date.
end_date	date	NOT NULL	
party_size	smallint	NOT NULL	

COLUMN	TYPE	CONSTRAINTS	NOTES
adults	smallint	NOT NULL	
children	smallint	NOT NULL	
trip_type	text	NOT NULL · one of solo, couple, family, business, friends, senior, backpacker	
is_group_trip	bool	NOT NULL	PS-11 / PS-08 filter.
status	text	NOT NULL · one of draft, planning, confirmed, in_progress, completed, cancelled	
home_currency	char(3)	FK → currencies.iso4217 · NOT NULL	
notes	text		
created_at	timestamptz	NOT NULL	
updated_at	timestamptz	NOT NULL	

user_preferences

Identity & preference · 1,200 rows · IDs start prf_

Explicit preference signal. PS-04's language-preference requirement reads from here.

COLUMN	TYPE	CONSTRAINTS	NOTES
preference_id	text	PK	prf_ prefixed.
user_id	text	FK → users.user_id · UNIQUE · NOT NULL	One row per user.
preferred_languages	text	NOT NULL	Comma-separated BCP-47 tags, most-preferred first.
guide_language	text	FK → languages.bcp47	PS-04 — preferred language for guide/tour delivery.
interests	text	NOT NULL	Comma-separated category codes.
dietary_flags	text		vegetarian
accessibility_needs	text		step_free
preferred_currency	char(3)	FK → currencies.iso4217 · NOT NULL	
max_daily_budget	decimal(12,2)		decimal, paired with currency below.
max_daily_budget_currency	char(3)	FK → currencies.iso4217	

COLUMN	TYPE	CONSTRAINTS	NOTES
pace	text	NOT NULL	relaxed
updated_at	timestampz	NOT NULL	

hotel_media

Supply & catalogue · 1,500 rows · IDs start `hmd_`

Image references with role and alt text. PS-05 uses hero/room rows as the fallback when no XR scene exists.

COLUMN	TYPE	CONSTRAINTS	NOTES
media_id	text	PK	hmd_ prefixed.
hotel_id	text	FK → <code>hotels.hotel_id</code> · NOT NULL	
room_type_id	text	FK → <code>hotel_room_types.room_type_id</code>	Null for property-level media.
media_role	text	NOT NULL · one of <code>hero</code> , <code>room</code> , <code>lobby</code> , <code>exterior</code> , <code>dining</code> , <code>pool</code> , <code>bathroom</code> , <code>view</code> , <code>landmark</code> , <code>menu</code> , <code>sign</code> , <code>receipt</code>	
file_path	text	NOT NULL	Relative path inside the media pack.
alt_text	text	NOT NULL	Accessibility and multimodal grounding.
width_px	int	NOT NULL	
height_px	int	NOT NULL	
sort_order	smallint	NOT NULL	
updated_at	timestampz	NOT NULL	

itineraries

Trip & itinerary · 803 rows · IDs start `itn_`

A versioned plan belonging to a trip. version is what makes PS-11's conflict handling tractable.

COLUMN	TYPE	CONSTRAINTS	NOTES
itinerary_id	text	PK	itn_ prefixed.
trip_id	text	FK → <code>trips.trip_id</code> · NOT NULL	
name	text	NOT NULL	
version	int	NOT NULL	Monotonic per itinerary.

COLUMN	TYPE	CONSTRAINTS	NOTES
is_active	bool	NOT NULL	Exactly one active version per trip.
generated_by	text	NOT NULL · one of user , ai_planner , optimizer , agent , vote , import	Which subsystem produced this version.
total_cost	decimal(12,2)	NOT NULL	sum of item costs, half-up at the end.
currency	char(3)	FK → currencies.iso4217 · NOT NULL	
total_duration_minutes	int	NOT NULL	
total_carbon_kg	decimal(10,3)	NOT NULL	APS-09 objective.
optimizer_weights	text		JSON string: {cost, time, carbon} weights that produced it.
status	text	NOT NULL · one of active , inactive , archived , draft	
created_at	timestampz	NOT NULL	
updated_at	timestampz	NOT NULL	

itinerary_items

Trip & itinerary · 8,583 rows · IDs start itm_

The atom of the portal, and the single most-shared object across the thirteen builds. If a team implements one shared shape, this is it.

COLUMN	TYPE	CONSTRAINTS	NOTES
item_id	text	PK	itm_ prefixed.
itinerary_id	text	FK → itineraries.itinerary_id · NOT NULL	
day_index	smallint	NOT NULL	1-based.
sort_order	smallint	NOT NULL	Order within the day.
starts_at	timestampz		Rule R4: offset-carrying; null for unscheduled items.
ends_at	timestampz		
item_type	text	NOT NULL · one of hotel , flight , poi , package , guide , transfer , meal , free	

COLUMN	TYPE	CONSTRAINTS	NOTES
entity_type	text	one of hotel, room_type, rate_plan, flight, flight_fare, poi, package, package_component, guide, transfer, event, xr_scene	Polymorphic supply reference.
entity_id	text		Canonical ID of the referenced supply row.
title	text	NOT NULL	
cost	decimal(12,2)	NOT NULL	D3.
currency	char(3)	FK → currencies.iso4217 · NOT NULL	
carbon_kg	decimal(8,3)	NOT NULL	
duration_minutes	int	NOT NULL	
source	text	NOT NULL · one of user, ai_planner, optimizer, agent, vote, import	user
explanation	text		Where APS-04 and PS-01 surface reasoning without a parallel structure.
locked	bool	NOT NULL	APS-09 hard constraint: must-see items cannot be dropped.
status	text	NOT NULL · one of proposed, confirmed, removed, replaced	Rule R8: removed items stay visible.
created_at	timestampz	NOT NULL	
updated_at	timestampz	NOT NULL	

bookings

Booking & money · 1,996 rows · IDs start bkg_

The order header — where inventory, payment and itinerary meet. Idempotency key is mandatory, not optional.

COLUMN	TYPE	CONSTRAINTS	NOTES
booking_id	text	PK	bkg_ prefixed.
user_id	text	FK → users.user_id · NOT NULL	
trip_id	text	FK → trips.trip_id	Null for standalone bookings.
itinerary_id	text	FK → itineraries.itinerary_id	

COLUMN	TYPE	CONSTRAINTS	NOTES
booking_reference	text	UNIQUE · NOT NULL	Human-quotable 6-char reference.
channel	text	NOT NULL · one of web, mobile_app, partner, call_centre, agent	
total_amount	decimal(12,2)	NOT NULL	must equal the sum of booking_items.
currency	char(3)	FK → currencies.iso4217 · NOT NULL	
tax_amount	decimal(12,2)	NOT NULL	
idempotency_key	text	UNIQUE · NOT NULL	APS-05 — retried requests resolve to this same row.
status	text	NOT NULL · one of pending, confirmed, partially_confirmed, cancelled, failed, refunded	
confirmed_at	timestampz		
cancelled_at	timestampz		
cancellation_reason	text		
created_at	timestampz	NOT NULL	
updated_at	timestampz	NOT NULL	

The rules that apply to these fields

#	RULE
R1	Additive only. Add columns, tables and stores freely. Never rename, drop or repurpose a field that came with the data.
R2	IDs are opaque prefixed strings — <code>htl_a91f3c</code> . Never integers, never parsed for meaning.
R3	Money is a pair: a 2-place decimal plus an ISO-4217 currency code. Never a float.
R4	Time is ISO-8601 with an offset. <code>_at</code> fields carry an offset; <code>_date</code> fields have no zone.
R5	Enums are lowercase snake_case and the legal values are in <code>data/enums.json</code> .
R6	Language is a BCP-47 tag — <code>ta</code> , not "Tamil".
R7	Geography is WGS-84 to 6 decimal places, <code>lat</code> and <code>lng</code> together or not at all.
R8	Nothing is hard-deleted. Rows carry <code>status</code> and <code>updated_at</code> .

Add whatever you like beside these fields — new columns, new tables, your own vector store, your own services. That is the point of R1. What you must not do is rename or re-key the fields that came with the data, because that is what would stop sixteen independent builds being put together afterwards.

`data/WORKING_WITH_THE_DATA.md` has the loading instructions, including how to read money without corrupting it. `tools/validate_conformance.py` tells you in thirty seconds whether you are still conformant.